

NCEI Five-Year Roadmap: FY 2020–2025



National Oceanic and Atmospheric Administration | National Centers for Environmental Information

Letter from the Director

For the first time since NCEI formed, we are releasing an organizational Roadmap of where we want to be in the next five years. NCEI's mission has never been more relevant than it is today. The rate of environmental change continues at a pace that makes NCEI's authoritative and timely environmental information essential for the prosperity and protection of our economy and society. To continue its unparalleled scientific service to the nation, NCEI must evolve our approaches to achieving our mission to adapt to unprecedented and disruptive changes in data sources and volumes, information technology (IT), and user needs.

We are setting this stretch aspiration for our organization, requiring us to leave our comfort zones, because there is an undeniable urgency for us to do so. The Roadmap will help us prioritize new initiatives across NCEI and inform which activities will need to be sunsetted as we strive to reach our aspiration. We must build these foundational capabilities if we want to take ownership in *NESDIS Reimagined*, support NOAA, and transform into the NCEI of the future.

The people of NCEI are our biggest asset, and I continue to be inspired by the motivation, commitment, and expertise of our entire workforce. One of the drivers to create the NCEI Roadmap came from input received from employee organizational health surveys conducted in 2018 and 2019. Your responses underscored the need for clear communication of where we want to go in the future and what needs to be done to get there. I am confident that together we will realize the ambitious goals set forth in this plan. I believe that by pursuing and investing in these areas, we will become a stronger organization and be equipped to face the challenges and welcome the opportunities of the future.



The NCEI Roadmap

The aspirational goal of our Roadmap is:

NCEI will meet 100% of NOAA mission requirements for retrospective environmental information by 2025.

We will achieve this goal by focusing on four thematic areas with specific outcomes and strategies to achieve those outcomes:

- People and Culture
- Scientific and Data Stewardship
- User Engagement and Information Services
- Transformative Technologies

Why This Direction?

We need to achieve the full potential of NCEI to realize the aspirational goal, which requires integration—an overarching theme throughout the Roadmap. We have an unprecedented opportunity to create significantly more value than the individual bytes of data in our archive. The Nation’s need for trusted and authoritative environmental information has never been greater

than it is today. NCEI is uniquely positioned to meet NOAA’s and the Nation’s need for environmental data and information, and our scientists are poised to use this wealth of data to unlock further insights about our Earth system.

The increasing demands from decision makers, and the scale and complexity of the earth system information, require NCEI to steward and provide access to increasingly large datasets, pioneer new products and services, and continuously evaluate existing ones to meet NOAA and our Nation’s evolving needs. And, we do all of this while holding an unwavering commitment to the highest standards of integrity.

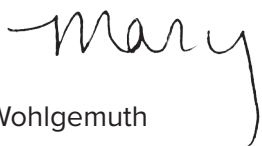
We must be prepared to bring our latest science and information products to the world, and this Roadmap is focused on collaboration and relationship building to improve our foundational capabilities related to our data stewardship and IT infrastructure so that we are ready for the future. These strategies will enable us to both leverage the full potential of NOAA’s mission data requirements for new applied science, and also bring us in line with current technology standards and address our long-standing technical debt. Our goal is to invest now in our stewardship capabilities so that we can bring to bear amazing new science advancements to NOAA and our partners in the future.

How Will We Use This Document?

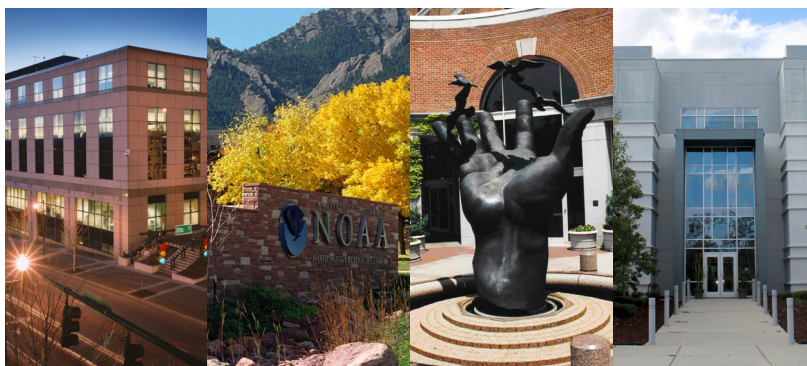
This document is not meant to be a point-by-point implementation plan but a guide that will inform our strategic direction as we navigate the next five years, and it will serve as the basis for measuring our success. We will use the Roadmap to drive annual operating plans, update our Organizational Health Action Plan, make resource decisions, align Federal Performance plans, and guide our behavior. Our success depends on everyone contributing to advancing NCEI towards this five-year vision. As you read through our new Roadmap, keep in mind what it is trying to accomplish: to establish common goals, to serve as a guide for prioritization, and to align the organization around our aspiration.

Our Roadmap is firmly aligned with *NESDIS Reimagined*, which is transforming the way our parent organization operates. The NESDIS Strategic Implementation Plan (SIP), which is similar to our Roadmap, lays out the NESDIS organizational priorities over the next five years. Our aspiration and Roadmap naturally flow into *NESDIS Reimagined* and will help us tell our story in a way that will resonate with our stakeholders and parent organizations.

Respectfully,



Mary Wohlgemuth



NESDIS Strategy

The [NESDIS Strategic Plan](#), supported by the [NESDIS Reimagined effort](#), is a new vision for NESDIS that will enable us to maintain our leading role in the field of environmental observation and understanding in a rapidly changing world. The goals set forth in the Strategic Plan and *NESDIS Reimagined*, our NCEI's goals, and this Roadmap lay out NCEI's contributions to our shared vision and mission. The *Reimagined* effort is happening because of an ever-evolving environmental observation landscape and the need for a better working environment and culture within NESDIS. The organizational health surveys that were conducted in 2018 revealed that NCEI needs clearer direction from leadership on how we support the NESDIS Strategy. The survey indicated the workforce is unsure of how NCEI's work contributes to achieving the NESDIS vision, needs direction to translate the mission into actionable goals, and requires clarity on execution priorities. This Roadmap for NCEI is meant to provide guidance each year as we develop our annual operating plan. It ties into the NESDIS Strategic Implementation Plan and gives our organization a Roadmap to contribute to the NESDIS Vision, Mission, and Aspiration.

NESDIS Vision:

"Expand understanding of our dynamic planet as the trusted source of environmental data."

NESDIS Mission:

"Provide secure and timely access to global environmental data and information from satellites and other sources to both promote and protect the Nation's environment, security, economy and quality of life."

NESDIS Aspiration:

"Provide a truly integrated digital understanding of our Earth's environment that can evolve quickly to meet changing user expectations by leveraging our own capabilities and partnerships."

NESDIS has also outlined Operating Principles, released In December 2018 by Dr. Stephen Volz, Assistant Administrator for Satellite and Information Services. These are instrumental features of a healthy organization, and the NCEI Values encompass these operating principles.



NCEI Values

NCEI will meet its aspirational goal set forth in this Roadmap and the NESDIS and NOAA mission by relying on building our culture around three values that encompass the NESDIS Operating Principles: **Courage, Agility, and Teamwork**. Our values serve to guide employee behavior at all levels and provide clarity and alignment across the organization. Clarity around how to interact and operate provides a decision standard for employees, reduces the need for micromanagement, and frees up supervisors to focus on leadership, management, and developing their people. This clarity also helps to attract and retain employees and serves to embed and strengthen the culture we are trying to build. The common vocabulary that flows from these values serves as a shortcut to referring to the way we need to interact, operate, and make decisions along the lines of the NESDIS Operating Principles.

Courage – *We empower each other to choose courage over comfort to do what is best for the organization.*

- We operate with personal and professional integrity and are empowered to risk failure, whether it is in resolving a conflict by initiating a difficult conversation or by acting boldly to devise and implement a new technological solution.
- We are empowered to act in the best interest of NCEI by making the tough decisions and taking the time we need to “get things right.”

Agility – *We constantly assess and adapt what we are doing and how we are doing it.*

- We work within a learning framework, taking risks and learning from failure to power innovation and to ensure constant incremental improvement in what we do and how we interact and operate together.
- We ensure constant feedback and interaction with internal and external stakeholders and clearly articulate our priorities so that people at all levels of the organization know what activities need to be accomplished.
- We communicate early and often about process or people challenges that may delay or derail our ability to meet our commitments, and we work together as an organization to address them.

Teamwork – *We work together to build cohesive, highly functioning teams.*

- We all have a sense of ownership and responsibility for our aspirational goal, are empowered to make decisions, and are clear on how we contribute to the success of the team.
- We are energized to come to work, the inherent value of each person is recognized, and everyone feels seen, heard, valued, and is treated with dignity and respect.
- We encourage team members to focus on results, to be vulnerable and genuine with each other, engage in passionate unfiltered debate around issues, cultivate clarity and buy-in around decisions, and hold each other accountable.

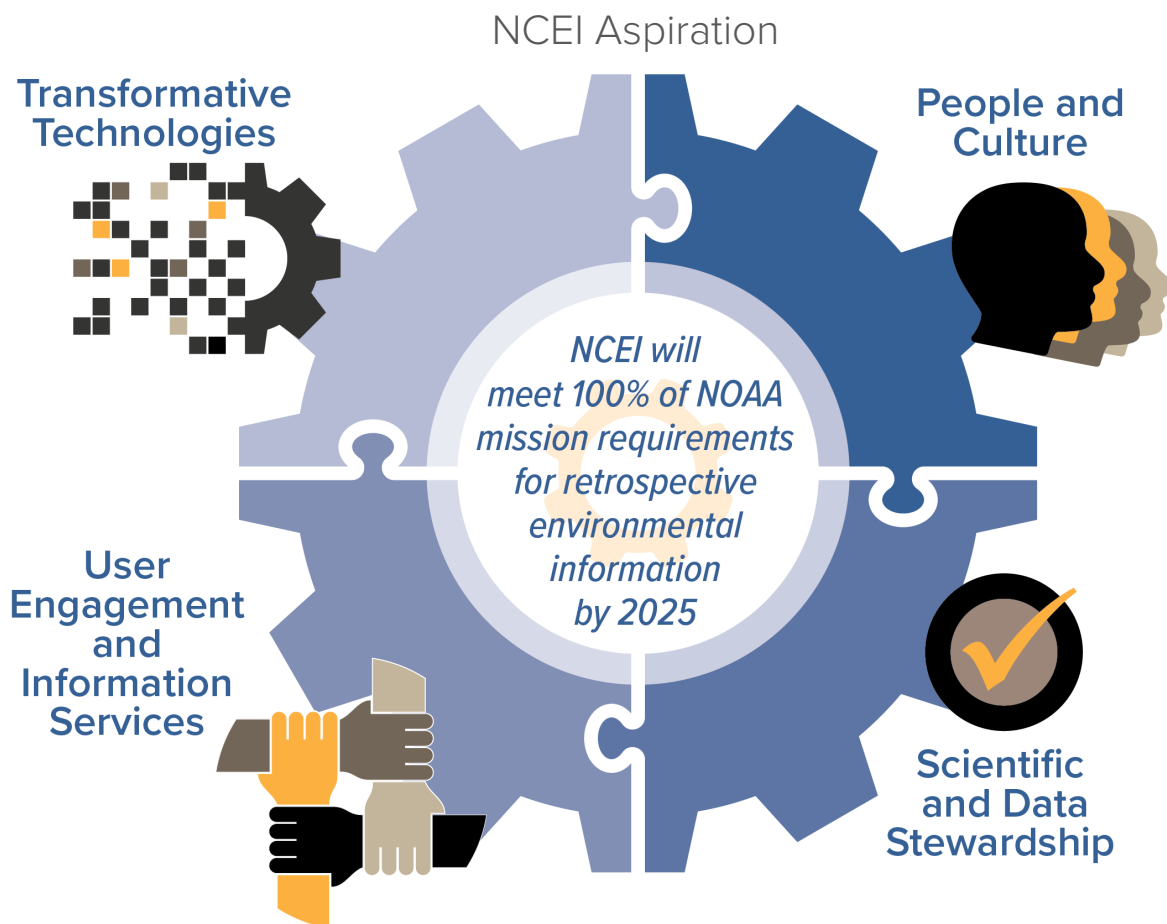


NCEI Aspiration

NOAA and its partners spend billions of dollars each year to collect high-quality environmental data; and it is the mission of NCEI to ensure these data are not just preserved, but that their utility to the Nation is maximized. NOAA and NESDIS have a long history of providing free and open access to massive volumes of data, preserving that data for perpetuity, and having scientific expertise to create high-value environmental intelligence. Guided by legal and administrative mandates and user needs, NCEI has scientific and data expertise to produce environmental intelligence that enables individuals, businesses, and governments to make informed decisions. It does this, in part, by providing science-based products and services that multiply the value of archived data and information.

Our internal NOAA stakeholders place mission requirements on NCEI across the tiers of stewardship, from basic archiving of retrospective data to higher-level scientific data products and assessments. Some of these requirements are met with NCEI base funding, while others are partner funded. Time and energy will be required to understand the full suite of NOAA's mission requirements by utilizing cross-NOAA Councils and Boards, such as the Environmental Data Management Council, among others.

Over the next five years, this aspirational statement will drive NCEI to be able to steward all NOAA data and focus our scientific efforts on advancing the NOAA mission. This aspirational statement is meant to build foundational capabilities that will allow us to propel our organization forward and rise to the next level, including establishing clear and transparent business practices for our partners and fully leveraging the emerging enterprise capabilities that are built across the agency to move NOAA data to the Cloud.



Tiers of Data Stewardship



NCEI data are fundamental to humans' understanding of Earth's environment, developing informed policy, and driving billions of dollars of commercial activity. In recent years, however, NCEI's approaches to achieving the mission have faced increasing strain driven by unprecedented and disruptive changes in data sources and volumes, information technology (IT), and user needs. To continue its unparalleled service to the Nation and world, NCEI must now adapt to these realities by making its own fundamental changes. This Roadmap describes the strategic goals that together lay the path for a modern, efficient, and scalable organization and national asset that continues to facilitate U.S. economic growth, sound decisions and policies, and an increasingly informed population throughout the coming decades.

Strengthening the connections between our Science, Service, and Stewardship enterprises will be essential to maximize the value of environmental information to the Nation. We will realize our aspiration by focusing on the following four thematic areas that are key for our organization:

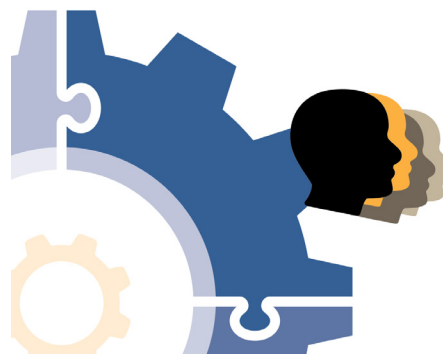
1. People and Culture
2. Scientific and Data Stewardship
3. User Engagement and Information Services
4. Transformative Technologies

NCEI's thematic areas are interdependent, and integration across all of them is essential for our collective success. Each thematic area will require support from all of NCEI's Centers and Divisions to be achieved. As a use-inspired science and service-based organization, our assessment, monitoring, and synthesis activities are built using data that are stewarded with complete transparency and provenance. NCEI's science and data stewardship are the basis for all of NCEI's use-inspired products and environmental information that are utilized by our stakeholders. And to ensure that NCEI's science and stewardship are use-inspired, they must be driven by user requirements. None of this can be achieved without a hybrid workforce of talented, engaged, and motivated people who have the right tools to do their jobs effectively.

NCEI Thematic Areas, Outcomes, and Strategies

People and Culture

People and Culture is about intentionally being efficient and effective in the way we do business, serve the public, and work together. NCEI will focus strategically not only on what we will do to achieve our Science, Service, and Stewardship mission, but also on how we will work together as oneNCEI in the process. NCEI will work to create an environment where people feel empowered to be themselves and fully contribute by embedding our values of **Courage**, **Agility**, and **Teamwork** into our culture. These values will serve as guiding principles for how we interact and operate to achieve our collective goals.



NCEI will build upon our shared commitment to the highest standard of integrity to create a culture of highly functioning, agile teams that can effectively respond to the changing needs of our internal and external stakeholders. NCEI will create a culture that naturally attracts and retains an engaged and productive workforce with a diversity of skills, expertise, and perspectives reflecting the communities we serve. The NCEI Roadmap lays out the strategic direction of the organization with specific outcomes and strategies to reach our goals. Other NCEI implementation plans, like the Organizational Health Action Plan, align with the outcomes and strategies laid out in the Roadmap, and the documents will work in concert to accomplish our People and Culture goals.

People and Culture Outcome 1: *NCEI's values are embedded in the culture so that all employees are empowered to fully contribute as oneNCEI to meet our organizational goals.*

Strategy 1: Foster effective communication across NCEI to build relationships, improve trust, connections, and team dynamics.

Strategy 2: Continuously evaluate organization health and adjust implementation strategies based on feedback to enable employees to be authentic, to fully contribute, and to establish high levels of trust and connection.

Strategy 3: Appropriately recognize and reward all employees for professional achievements and contributions to advancing NCEI's aspirational values.

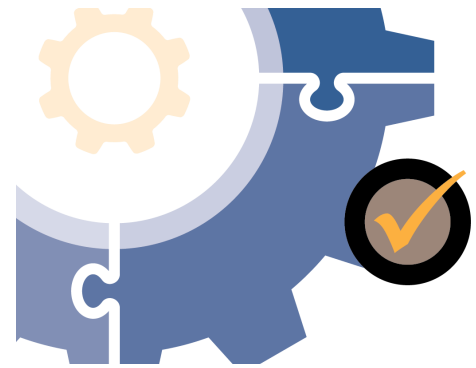
People and Culture Outcome 2: *NCEI's workforce is positioned to leverage transformative technologies, emerging opportunities, and evolving organizational priorities.*

Strategy 1: Shape the workforce to meet changing organizational priorities.

Strategy 2: Develop the workforce for succession opportunities and ensure employees have access to training to meet NCEI's evolving technical and administrative needs.

Scientific and Data Stewardship

NCEI is a science-based service organization that produces environmental information to enable individuals, businesses, and governments to make informed decisions, and provides the foundation for more tailored decision-support services to be developed and delivered by the public and private sector. By integrating across the science disciplines within NCEI, we unlock timely, actionable, and reliable environmental information about our Earth system. NCEI's scientific success is also dependent on strategic and sustainable national and international partnerships. NCEI's science enterprise is built on our mandate to archive the growing amount of environmental data and information generated by NOAA. The science enterprise is use-inspired and responsive to input gathered through NCEI's user engagement efforts.



For the Nation to fully realize the long-term value of NOAA's significant investment in collecting observations, generating model outputs, and deriving products for public safety and commerce, NCEI must first effectively steward the data so information can be easily found, accessed, and used by others. NCEI's activities must also provide our science enterprise with access to data that meet the highest standards of transparency, integrity, quality, and provenance. To achieve these results, NCEI maintains the systems, policies, and procedures that enable long-term information stewardship in a cost-effective, efficient, and reliable manner.

Scientific and Data Stewardship Outcome 1: *NCEI's enterprise science and stewardship capabilities effectively scale and serve NOAA's evolving needs.*

Strategy 1: Define and communicate NCEI's science and stewardship business model to provide transparency and consistency for our stakeholders.

Strategy 2: Actively engage in shaping and aligning with the NESDIS user engagement, requirements, and configuration management processes to inform responsive decision making.

Strategy 3: Inform and leverage NOAA and NESDIS enterprise architecture to achieve scalable and sustainable NCEI science and stewardship functionality.

Scientific and Data Stewardship Outcome 2: *NCEI leads NESDIS in being a trusted data provider.*

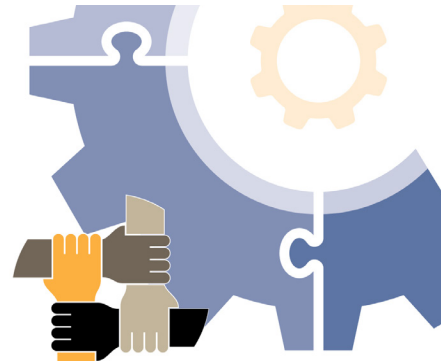
Strategy 1: Develop and implement common data management and quality guidelines across the NCEI portfolio.

Strategy 2: Expand data quality standard processes to the NESDIS enterprise.

Strategy 3: Ensure continued curation and preservation of current and future data holdings.

User Engagement and Information Services

Decision makers around the Nation are increasingly recognizing the value of using NCEI's environmental data to better manage risk and exploit new opportunities. NCEI will continue to strengthen and balance its portfolio of science-based products and services to meet the needs of its stakeholders and improve the delivery of this information. An important mechanism through which NCEI services are provided and advanced is a network of regional and sector-focused channels. In addition, NCEI's information would not fully be realized without our partners who help users access and understand NCEI's assets. Those who use NCEI data, such as the private sector, improve our assets by suggesting value-added services. Our user engagement and information services activities will be consistent with NESDIS enterprise-wide user engagement and integrated program development. NESDIS's path forward on user engagement will provide the framework for our organization moving forward in this thematic area.



User Engagement and Information Services Outcome 1: *NCEI engages with users to understand and manage 100% of NOAA requirements for retrospective environmental information.*

Strategy 1: Collaborate with users to understand, document, and share their needs for existing and future products and services throughout the full life cycle.

Strategy 2: Establish NOAA mission requirements management process for NCEI including gathering and validating requirements as well as evaluating and implementing those requirements. Use relevant councils to aid in prioritization and adjudication.

Strategy 3: Consistent with NESDIS enterprise-wide development, align user engagement and information services goals with science and data stewardship goals.

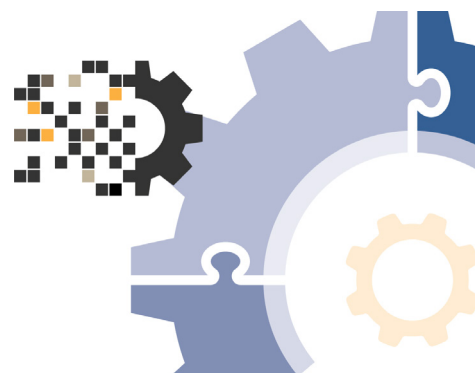
User Engagement and Information Services Outcome 2: *NCEI innovates how it provides access to and usability of environmental data to meet user requirements.*

Strategy 1: Exploit new technology and enterprise systems to assist users in finding and using data efficiently.

Strategy 2: Improve communications and feedback mechanisms with customers on access and information systems.

Transformative Technologies

In November 2019, NOAA announced [four new strategies](#) to dramatically expand the agency's application of four emerging science and technology focus areas—NOAA Unmanned Systems, Artificial Intelligence, 'Omics, and the Cloud—to guide transformative advancements in the quality and timeliness of NOAA science, products, and services. These strategies depend on partnerships in science and technology, and engage the community to explore the unknown ocean, collect data in new and innovative ways, advance science and our understanding of the Earth system, and promote new technologies. All of these strategies impact NCEI's mission by bringing in new data that will need stewardship, creating opportunities for scientific advancements using NCEI data and increasing the value of the data and NCEI's ability to serve it to the user community. To meet the future mission requirements for NOAA, NCEI must embrace new technologies that hold incredible promise to solve our greatest challenges.



Transformative Technology Outcome 1: *100% of NCEI's data is accessible through the Cloud to enable a truly integrated digital understanding of our Earth's environment.*

Strategy 1: Support the development of NESDIS Cloud Framework by continuing engagement with NESDIS Cloud Pilot, Analysis of Alternatives, and Concept of Operations (CONOPS).

Strategy 2: Perform analysis of existing data and applications to prepare for eventual move into the NESDIS Cloud Framework.

Strategy 3: Transition internal culture to adopt Agile Framework and DevSecOps processes in line with NESDIS Cloud Framework.

Strategy 4: Leverage relationships with affiliates and partners to accelerate Cloud migration and tool development.

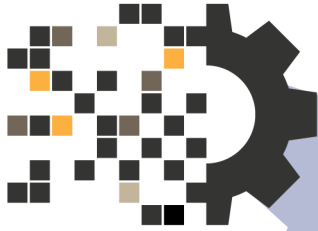
Transformative Technology Outcome 2: *NCEI operates with maximum operational and organizational efficiency to optimize the use of resources.*

Strategy 1: Invest in and implement transformative business technologies (e.g., financial management, document management) that automate processes and maximize organizational efficiency.

Strategy 2: Exploit Artificial Intelligence (AI) Machine Learning (ML) technologies to develop transformation capabilities for data stewardship and scientific analysis.

NCEI Aspiration

Transformative
Technologies



People and
Culture



*NCEI will
meet 100% of NOAA
mission requirements
for retrospective
environmental
information
by 2025*

User
Engagement
and
Information
Services



Scientific
and Data
Stewardship

